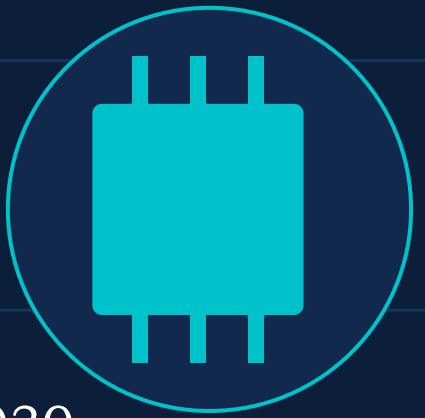
SEMINAR | ECQ 413

Advanced Packaging Technologies for VLSI



Guided by: **Prof. Jithesh C P**
Assistant professor
Department of ECE,
GEC Palakkad

Presented by: **Gopika PS**
PKD23EC030

Contents

1. Introduction
2. Why Packaging Matters in VLSI Design
3. Evolution of VLSI Packaging
4. Comparison of packaging Technologies
5. 2.5D Integration — Principle, Advantages & Challenges
6. 3D Integration — Principle, Advantages & Challenges
7. Wafer-Level & Fan-Out Wafer-Level Packaging (WLP / FOWLP)
8. Implications for Design Tools: Signal Integrity, Power & Thermal
9. System-Level SImulation & Verification Methodologies
10. Conclusion & Future Directions

Introduction

- Transistor scaling is nearing its physical limits.
- Packaging has shifted from a passive protective role to an active performance enabler.
- Advanced packaging now drives gains in speed, power efficiency, and form factor.
- **This seminar covers the key packaging technologies shaping modern VLSI systems.**

Why Packaging Matters in VLSI Design

1

Electrical Interconnection

Routes signals between die and external connectors; advanced packaging improves signal integrity and power distribution.

2

Thermal Management

Heat sinks, thermal vias & TIMs dissipate heat as devices shrink and power density rises.

3

Form Factor

Compact, miniaturized packages enable smaller, lighter, more portable electronics.

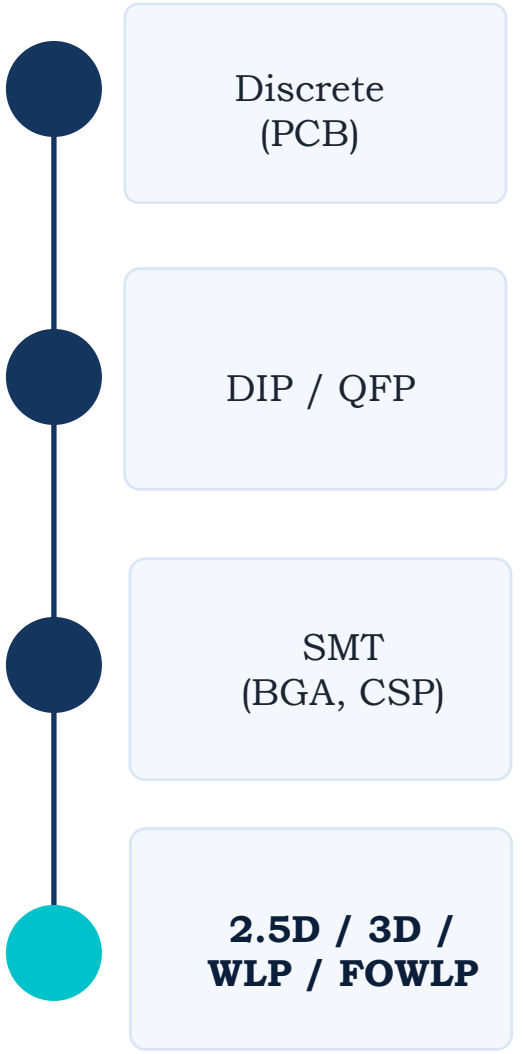
4

System Integration

Serves as the bridge that lets ICs function reliably within larger electronic systems.

Evolution of VLSI Packaging

- VLSI packaging encloses ICs and connects them to external circuitry a discipline that has evolved alongside chip technology.
- **Discrete era:** individual components wired together on PCBs.
- **Integrated packaging:** DIP (Dual Inline Package) and QFP (Quad Flat Package) multiple ICs in one package, smaller footprint, better reliability.
- **SMT (Surface-Mount Technology):** BGA (Ball Grid Array) and CSP (Chip Scale Package) no wire bonding, automated assembly, higher density.
- **Today:** 2.5D, 3D Integration, WLP, and FOWLP enabling miniaturization and heterogeneous integration.



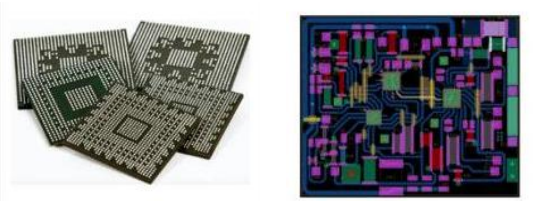


Fig 1:PCB (From Google)

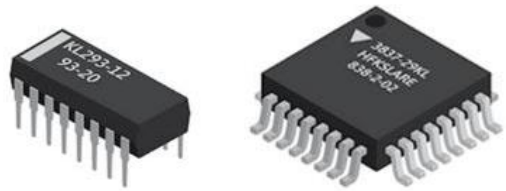


Fig 2:DIP & QFP(From Google)

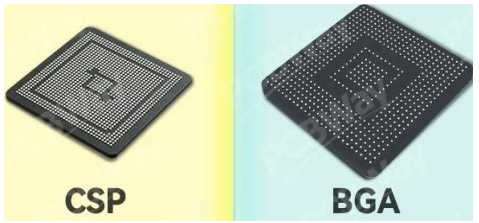


Fig 3 :CSP & BGA(From Google)

Motivation for Advanced Packaging

Overcoming the limits of conventional packaging

Traditional packaging limits interconnect density, power efficiency, and system integration. 2.5D, 3D Integration, WLP, and FOWLP unlock heterogeneous integration of ICs, sensors, and MEMS in a single package.

Parameter	2D Integration	2.5D Integration	3D Integration	WLP	FOWLP
Integration Density	Low	Medium–High	High	Medium	High
Electrical Performance	Good	Very Good	Excellent	Good	Excellent
Thermal Management	Poor	Moderate	Challenging	Moderate	Good
Cost Efficiency	Low	Moderate	Moderate	High	Moderate
Power Efficiency	Low	High	Very high	Moderate	High

Table 1: Comparison between advanced packaging technologies

2.5D Integration — Principle & Mechanism

- **Multiple dies are placed side-by-side (not stacked) on a common silicon interposer.**
- Dies are mounted on the interposer using micro-bumps.
- The interposer carries fine-pitch redistribution layers (RDL) that connect the dies to each other.
- Through-Silicon Vias (TSVs) in the interposer connect it down to the package substrate below.
- **Applications: High-Bandwidth Memory (HBM) systems, GPUs and AI accelerators requiring fast memory access.**

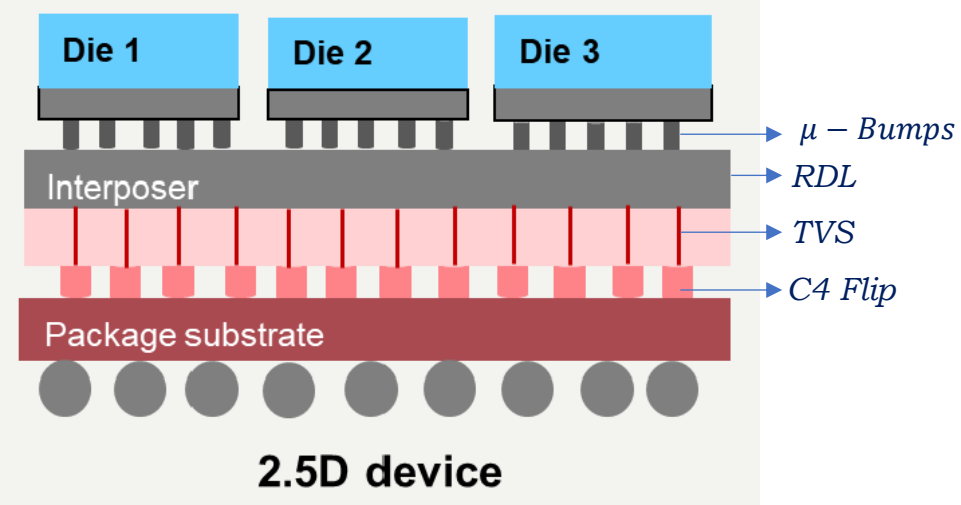


Fig 4:2.5 device (from Google)

2.5D Integration — Advantages & Challenges

ADVANTAGES

- High-bandwidth, low-latency communication between dies via the interposer
- Enables heterogeneous integration — combining logic, memory, and analog components in one package
- Lower power consumption due to shorter interconnect paths
- Better performance than traditional side-by-side packaging on a PCB

CHALLENGES

- Interposer fabrication adds cost to the overall package
- Still limited by 2D die placement — does not reach the density of full 3D stacking

3D Integration — Principle & Mechanism

- Multiple semiconductor dies are stacked vertically to form a three-dimensional structure.
- Through-Silicon Vias (TSVs) — vertical interconnects penetrating the silicon substrate — provide electrical connections between stacked layers.
- TSVs enable efficient signal routing between dies without relying solely on peripheral wire bonds.

- **Applications: High-performance computing, memory stacking, and complex System-on-Chip (SoC) integration.**

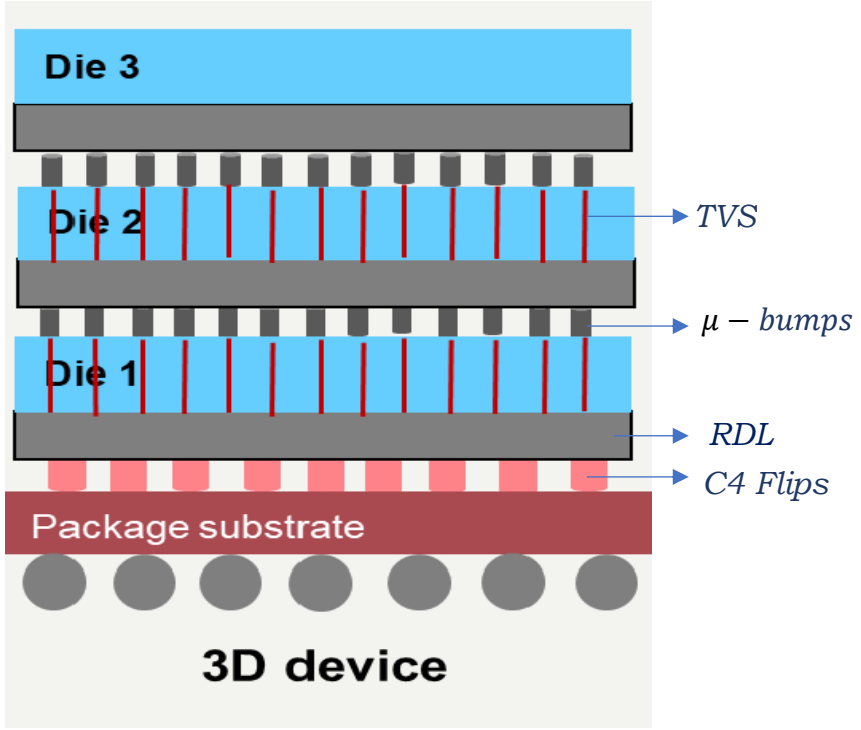


Fig 5: 3D device (from Google)

3D Integration — Advantages & Challenges

ADVANTAGES

- Increased integration density — stacking multiple dies in a compact footprint
- Improved performance — shorter interconnects, reduced parasitics, faster signaling
- Enhanced power efficiency — shorter paths and better thermal handling lower power consumption

CHALLENGES

- Thermal management — high power density in stacked dies risks overheating
- Manufacturing complexity — fabricating TSVs and aligning dies raises cost & yield risk
- Design complexity — new methodologies needed for routing, power, and thermal modeling across layers

Wafer-Level Packaging (WLP)

Packaging performed directly on the wafer, before dicing

- **Definition: The entire packaging process — bumping, redistribution, and testing — is carried out on the silicon wafer before it is diced into individual chips.**
- Redistribution layers (RDL) are built directly on the wafer surface to reroute connections.
- Solder bumps are added on the wafer itself.
- The wafer is diced into individual chips only after packaging is complete.

Applications: Mobile phones, wearables, and IoT devices requiring a small form factor.

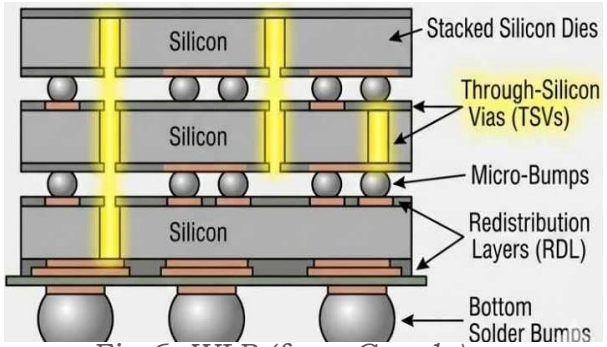


Fig 6: WLP (from Google)

BENEFITS

- Package size nearly equals die size — very compact
- Fewer processing steps → lower cost, high-volume friendly
- Shorter interconnects improve electrical performance

CHALLENGES

- Limited to smaller die sizes and lower pin counts
- Less flexibility for combining multiple dies in one package

Fan-Out Wafer-Level Packaging (FOWLP)

Interconnects fan out beyond the die edges — no substrate needed

- **Definition: The die is placed in a reconstituted wafer, letting interconnects extend beyond the original die edges — removing the need for a traditional substrate.**
- The die is embedded in a mold compound to form a new, larger reconstituted wafer.
- Redistribution layers are built over both the die and the surrounding molded area.
- The fan-out area allows more I/O connections than the die's original size would normally allow.

Applications: Smartphones, mobile processors, and compact high-performance IoT devices

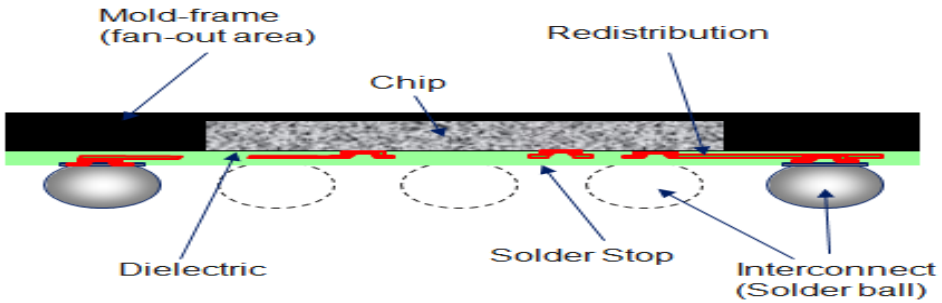


Fig 7: FOWLP(from Google)

BENEFITS

- Higher interconnect density than standard WLP
- No substrate required → thinner, lighter package
- Better electrical performance and thermal characteristics

CHALLENGES

- More complex manufacturing process than standard WLP
- Precise die placement in the mold compound is critical to yield

Design Tool Implications: Signal Integrity

Compact layouts and high-speed interconnects demand rigorous analysis

- Advanced packaging's dense layouts and fast interconnects can cause signal distortion, crosstalk, and electromagnetic interference (EMI).
- Designers rely on advanced simulation tools throughout the design flow to catch and mitigate these issues early.

1 EM Simulators

Analyze electromagnetic fields to predict signal behavior and optimize layouts against interference.

Eg; Ansys HFSS, Cadence Clarity, Keysight ADS (RFPro), Synopsys Totem

2 High-Speed Signal Integrity Analyzers

Analyze high-speed signal propagation and flag design changes to improve performance.

Eg; Cadence Sigrity X, Ansys HFSS, Synopsys PrimeSim, Keysight PathWave ADS

3 Crosstalk Analysis Tools

Assess coupling between adjacent traces to optimize routing and minimize interference.

Eg; Synopsys PrimeTime SI, Cadence Quantus Extraction Solution,

Ansys / HFSS

Fig 8:Ansys (from google)

cadence

Clarity 3D Solver

Fig 9:Cadence (from google)

SYNOPSYS®

Fig 10:Synopsys(from google)

Power Distribution & Thermal Management

- Higher power densities and thermal resistances in advanced packages demand sophisticated Power Delivery Network (PDN) analysis and thermal simulation.
- Thermal-aware routing algorithms minimize thermal gradients for uniform heat dissipation.
- Heat sinks, thermal vias, and Thermal Interface Materials (TIMs) enhance conductivity and reliability.

- **Thermal Analysis Tools:** Simulate heat generation, conduction & dissipation to locate hot spots.
- **Thermal-Aware Routing:** Optimize layout to spread heat evenly and improve reliability.
- **Thermal Enhancement Techniques:** Heat sinks, thermal vias, and TIMs(Thermal Interfacing Material) boost heat dissipation.

EXPERIMENTAL BENCHMARK

> 1 KW/cm²

power density dissipated with liquid microchannel cooling, at a max. temperature rise of only 80°C (Tuckerman & Pease heat-sink program, Stanford).

Microcapillary chip-attach interfaces achieved thermal conductance of 12–25 W/K·cm², validating liquid cooling as a practical thermal-management route for dense packages.

System-Level Simulation & Tool Enhancements

System-Level Simulation

- Co-simulation platforms model interactions between components, subsystems, and software layers.
- Virtual prototyping tools explore architecture trade-offs before physical implementation.

Design Tool Enhancements Needed

- 2.5D / 3D Integration Support — model interposers & TSVs, optimize routing, run thermal simulations
- Wafer-Level Packaging Tools — optimize die placement & redistribution layers (RDL)
- FOWLP Support — integrate heterogeneous dies, passives, and RDL designs

Bottom line:

Design automation must evolve so engineers can model, simulate, and optimize designs across 2.5D, 3D Integration, WLP, and FOWLP with confidence.

Verification & Validation Methodologies

Verification and validation in advanced packaging ensure that complex heterogeneous systems meet rigorous electrical, thermal, and mechanical performance standards while maintaining high manufacturing yield.

1 Advanced Simulation

Electrical, thermal & mechanical analyses verify functionality and performance under varied operating conditions.

Eg;ANSYS, COMSOL Multiphysics, Siemens Simcenter, Keysight PathWave

2 Design for Testability (DFT)

Ensures designs are testable and manufacturable — detecting faults during manufacturing and operation.

EDA Software: Synopsys, Cadence Design Systems
ATE Hardware: Teradyne, Advantest
Physical Inspection: KLA Corporation, Nordson Test & Inspection

3 Hardware Emulation & Prototyping

Hardware-accelerated platforms validate system functionality and specification compliance in real-world settings.

Eg; Synopsys ZeBu
Synopsys HAPS Series
Cadence Palladium System
Cadence Protium System
Siemens Veloce

Conclusion & Key Takeaways

1. 2.5D, 3D Integration, WLP, and FOWLP each unlock higher integration density, performance, and functionality — with distinct trade-offs in cost and thermal behavior.

2. Signal integrity, power distribution, and thermal management must be co-optimized as packaging complexity increases.

3. Design tools and methodologies must evolve — simulation, verification, and validation techniques tailored to advanced packaging are essential.

4. Success requires collaboration between design, packaging, and manufacturing teams, plus continuous upskilling of the VLSI design community.

5. Embracing AI-driven design automation and next-generation materials will dictate the next frontier of advanced packaging innovation.

Key References

[1] C. Bailey, “Reliability Challenges for Advanced Packaging,” in Proceedings of the IEEE 43rd VLSI Test Symposium (VTS), Tempe, AZ, USA, 2025.

[2] N. Sharma and H. Tyagi, “Advanced Packaging Technologies for VLSI: Implications for Design Tools and Methodologies,” Journal of Research in VLSI Design Tools and Technology, vol. 9, no. 1, pp. 15–26, Jan.–Apr. 2024.

[3] J. H. Lau, “Recent Advances and Trends in Advanced Packaging,” IEEE Transactions on Components, Packaging and Manufacturing Technology, vol. 12, no. 2, pp. 228–252, Feb. 2022.

[4] R. Guo and S. Li, “Fan-out wafer level packaging (FOWLP) for semiconductor and electronic applications,” Microelectronics Reliability, vol. 83, pp. 218–225, 2018.

[5] R. Ghodsi, S. A. R. Zaidi, and D. DiMase, “Advanced packaging techniques for VLSI applications,” IEEE Transactions on Components, Packaging and Manufacturing Technology, vol. 8, no. 11, pp. 1910–1918, 2018.